

COPPER ETHERNET & UTP CABLE QUICK REFERENCE

CCNA / CCNP / CCIE Network Engineer Specification & Field Cheat Sheet

ANSI/TIA-568 & ISO/IEC 11801 STANDARDS

TWISTED-PAIR CABLE STANDARDS & TECHNICAL SPECIFICATIONS (CAT1 TO CAT8.2)

CATEGORY	YEAR	IEEE STANDARDS	MAX SPEED	FREQUENCY	MAX REACH	SHIELDING (ISO/IEC)	PAIRS / CONNECT.	AWG GAUGE	PRIMARY FIELD APPLICATION & ENGINEERING NOTES
CAT 1	1985	POTS / ISDN	< 1 Mbps	0.4 MHz	Distance Varies	U/UTP Unshielded	2 Pairs / RJ11	22–24 AWG	Legacy analog telephone voice lines. Not rated for Ethernet data networks.
CAT 2	1989	Token Ring / Arcnet	4 Mbps	4 MHz	100 m	U/UTP Unshielded	4 Pairs / RJ11, RJ45	22–24 AWG	IBM Token Ring 4 Mbps & Arcnet systems. Obsolete in modern enterprise infrastructure.
CAT 3	1990	10BASE-T (802.3i)	10 Mbps	16 MHz	100 m	U/UTP Unshielded	4 Pairs / RJ45	24 AWG	First standard for 10BASE-T Ethernet (uses 2 pairs: 1,2 Tx, 3,6 Rx). Legacy VoIP telephony.
CAT 4	1992	Token Ring 16	16 Mbps	20 MHz	100 m	U/UTP Unshielded	4 Pairs / RJ45	22–24 AWG	Designed for 16 Mbps Token Ring networks. Extremely brief lifespan before CAT5 rollout. Obsolete.
CAT 5	1995	100BASE-TX (802.3u)	100 Mbps	100 MHz	100 m	U/UTP Unshielded	4 Pairs / RJ45	24 AWG	Fast Ethernet standard (uses 2 pairs: 1,2 Tx, 3,6 Rx). Superseded by CAT5e in 2001. Superseded.
CAT 5e	2001	1000BASE-T (802.3ab) 2.5GBASE-T (802.3bz)	1 Gbps 2.5 Gbps	100 MHz	100 m (1G) 100 m (2.5G)	U/UTP , F/UTP	4 Pairs / RJ45	24 AWG	PoE 802.3af/at Gigabit Ethernet across all 4 pairs (full-duplex). Adds Far-End Crosstalk (FEXT) specs. Minimum baseline for modern legacy installs.
CAT 6	2002	1000BASE-T (802.3ab) 10GBASE-T (802.3an)	1 Gbps 10 Gbps	250 MHz	100 m (1G) 37–55 m (10G)	U/UTP , F/UTP	4 Pairs / RJ45	23 AWG	PoE+ 802.3at Internal physical spline/separator to reduce NEXT. Supports 10GBASE-T up to 37m (high noise) or 55m (favorable Alien Crosstalk environment).
CAT 6A	2008	10GBASE-T (802.3an) 5GBASE-T (802.3bz)	10 Gbps	500 MHz	100 m	U/FTP , F/FTP , S/FTP	4 Pairs / RJ45	23 AWG	PoE++ 802.3bt Full 100m channel for 10G. Eliminates Alien Crosstalk (ANEXT). Recommended industry standard for high-density enterprise & WiFi 6/6E/7 AP drops.
CAT 7	2002	ISO Class F (10GBASE-T)	10 Gbps	600 MHz	100 m	S/FTP (Screened/Foiled)	4 Pairs / GG45, TERA, RJ45	22–23 AWG	ISO/IEC 11801 standard (NOT recognized by TIA/EIA). Requires individual foil pair shield + braid. Non-standard connectors required for full spec.
CAT 7A	2010	ISO Class F _a (10G / 25G short)	10 Gbps 25 Gbps (50m)	1000 MHz	100 m (10G) 50 m (25G)	S/FTP (Screened/Foiled)	4 Pairs / GG45, TERA	22–23 AWG	Enhanced Class F. ISO standard only (Not TIA recognized). Designed for high-frequency AV and industrial applications. High immunity to EMI.
CAT 8.1	2016	25GBASE-T / 40GBASE-T (IEEE 802.3bq)	25 Gbps 40 Gbps	2000 MHz (2 GHz)	30 m (2-conn channel)	F/UTP , S/FTP (Class I)	4 Pairs / RJ45	22–24 AWG	4UPoE 90W TIA-568-C.2-1 standard. Backward compatible RJ45. Built specifically for Data Center Top-of-Rack (ToR) and End-of-Row server connections.
CAT 8.2	2016	25GBASE-T / 40GBASE-T (IEEE 802.3bq)	25 Gbps 40 Gbps	2000 MHz (2 GHz)	30 m (2-conn channel)	F/FTP , S/FTP (Class II)	4 Pairs / TERA, GG45	22–24 AWG	ISO/IEC 11801-1 Class II standard. Non-RJ45 connector interface (TERA/GG45). Highest shielding spec for data centers with stringent noise requirements.

1. ISO/IEC 11801 Cable Shielding Designations [XX/YTP]

Format: [XX] / [Y] [TP] → XX = Outer Shield, Y = Pair Shielding, TP = Twisted Pair

3. Power over Ethernet (PoE) IEEE Standards Summary

PoE Standard	Type	PSE Power	PD Power	Pairs Used	Min Cable
802.3af (PoE)	Type 1	15.4 W	12.95 W	2 Pairs	Cat 3 / Cat 5

6. Senior Engineer Transmission Metrics & Testing Terms

- **NEXT (Near-End Crosstalk)**: Signal coupling between pair 1 and pair 2 measured at the transmitter end. Caused by untwisting pairs too far at RJ45 termination (>0.5 inch / 12mm max).

Code	Outer Shield (XX)	Pair Shielding (Y)	EMI Protection
U/UTP	None (Unshielded)	None (Unshielded)	Basic (Standard)
F/UTP	Foil Shield	None (Unshielded)	Moderate (RFI/EMI)
U/FTP	None	Foil Shield per pair	High (Prevents Crosstalk)
S/FTP	Overall Braided Mesh	Foil Shield per pair	Maximum (Industrial/DC)
SF/UTP	Braid + Foil Shield	None (Unshielded)	Very High Outer Noise

2. T568A vs T568B RJ45 Pinout Color Standards

Pin	Pair (568A)	T568A Color Wire	Pair (568B)	T568B Color Wire
1	3 Tx+	White/Green	2 Tx+	White/Orange
2	3 Tx-	Green	2 Tx-	Orange
3	2 Rx+	White/Orange	3 Rx+	White/Green
4	1 Bi-	Blue	1 Bi-	Blue
5	1 Bi+	White/Blue	1 Bi+	White/Blue
6	2 Rx-	Orange	3 Rx-	Green
7	4 Bi+	White/Brown	4 Bi+	White/Brown
8	4 Bi-	Brown	4 Bi-	Brown

* **Straight-Through:** Same end spec (A-A or B-B). **Crossover:** T568A on end 1, T568B on end 2 (Auto-MDIX renders crossover cables obsolete on modern switches).

PoE Standard	Type	PSE Power	PD Power	Pairs Used	Min Cable
802.3at (PoE+)	Type 2	30.0 W	25.5 W	2 Pairs	Cat 5e / Cat 6
802.3bt (PoE+ +)	Type 3	60.0 W	51.0 W	4 Pairs	Cat 6A recommended
802.3bt (4UPoE)	Type 4	90.0 W	71.3 W	4 Pairs	Cat 6A (23 AWG)

4. ANSI/TIA-568 Structured Cabling Rules & Limits

- **100m Total Channel Limit:** 90m Permanent Link (Solid core horizontal cable) + 10m total Patch Cords (Stranded wire: max 5m in closet + 5m at desk).
- **Conductor Gauge (AWG):** Solid copper 23 AWG (Cat6A) vs 24 AWG (Cat5e). Lower AWG = thicker copper, less DC resistance & heat in PoE bundles.
- **Bend Radius:** Minimum 4x cable outer diameter (UTP), 8x diameter for STP/ Cat6A/Cat8 under tension to prevent pair deformation.
- **Bundling Heating (PoE 802.3bt):** Large cable bundles carrying 60W–90W experience temperature rise; reduce bundle size to max 24 cables or derate length.
- **Grounding / Shield Bonding:** Shielded cables (STP/FTP) **MUST** be grounded at patch panel via bonded drain wire to building main earthing terminal. Unbound shield acts as antenna, increasing noise!

5. Network Topology & Cable Selection Rule of Thumb

- **Horizontal Drops to Workstation / AP:** Cat 6A U/FTP or F/UTP (Future-proof for 10GBASE-T & WiFi 6E/7 APs requiring 90W PoE++).
- **Data Center Top-of-Rack (ToR):** Cat 8.1 S/FTP for up to 30m 25G/40G copper link (or DAC/Twinax/MMF Fiber).
- **Industrial / High EMI Factory:** Cat 6A / Cat 7A S/FTP with M12 or shielded RJ45 industrial connectors.

- **FEXT / ELFEXT (Far-End Crosstalk):** Interference measured at the far receiver end of the cable link.
- **PS-NEXT (Power Sum NEXT):** Combined crosstalk effect on one pair from all other 3 pairs active simultaneously (critical for 1000BASE-T 4-pair operation).
- **ANEXT (Alien Crosstalk):** Signal coupling between *adjacent cables* in a bundle. Primary performance limiter for 10GBASE-T over Cat6 (solved in Cat6A by internal spline + pair shield spacing).
- **Return Loss (RL):** Signal reflections caused by impedance mismatches (e.g., kinks, harsh bends, cheap RJ45 connectors, bad crimps).
- **Propagation Delay & Skew:** Time difference between fastest and slowest pair to reach receiver. Max skew < 50ns per 100m.

7. Connector Compatibility Matrix

Connector	Form Factor	Max Cat	Notes / Backwards Compatibility
RJ11	6P2C / 6P4C	Cat 3	Voice lines / POTS. Fits physically into RJ45 jack but damages outer pins 1 & 8 if forced!
RJ45	8P8C	Cat 8.1	Universal Ethernet standard. Rated up to 2 GHz for Cat 8.1 channel.
GG45	ARJ45 format	Cat 7A / 8.2	ISO standard with extra contact pins in corners. Backwards compatible with RJ45 plugs.
TERA	Non-RJ style	Cat 7A / 8.2	Siemon proprietary ISO Class II connector. Quadrant pair separation. Not RJ45 compatible.

8. CCNA/CCNP Field Diagnostic & Fluke Test Checklist

[PASS] Wire Map: All 8 conductors pin-to-pin correct (No split pairs!) [PASS] Length: Total channel < 100m (Perm Link < 90m) [PASS] NEXT & PS-NEXT Margin: > 3.0 dB headroom [PASS] Return Loss Margin: > 2.5 dB [PASS] Shield Continuity: Drain wire grounded to patch panel